Siddhant Pathak

+65-86504682 | siddhantpathak2@gmail.com | siddhantpathakk.com | linkedin.com/in/siddhantpathakk

Education

Nanyang Technological University Aug 2020 – May 2024

Bachelor of Engineering in Computer Science (Elective Focus in Data Science and Artificial Intelligence)

- CGPA: 4.27/5.00 (Honours with Distinction)

Experience

Singapore Airlines Aug 2024 – Present

Data Scientist *Singapore*

- Re-architected the customer-facing Flight Recommender service, cutting infrastructure cost by ∼90% and end-to-end latency by 33% while preserving recommendation quality. Try it out here.
- Designed and shipped an internal LLM gateway powering 200+ AI applications enterprise-wide, standardising auth, rate-limiting, model routing, and observability across teams.
- Built a RAG-powered search engine on Agent 360, the B2B travel-trade portal, giving travel agents grounded, citation-backed answers over Singapore Airlines' policy and fare content.
- Delivered a natural-language interface for authoring and orchestrating agentic workflows, enabling non-engineers to compose multi-tool agents without writing code.
- Led deployment of open-source LLMs on cloud GPUs and on-prem Mac Studio clusters, owning capacity planning, vLLM serving, and inference cost/latency tuning.

Asurion Jan 2024 – Apr 2024

Data Science & GenAI Engineer (Intern)

- Built ML models for supply-chain optimisation and customer segmentation, and shipped a LangChain-based retrieval chatbot that accelerated support ticket resolution and reduced churn.

Glance, InMobi May 2023 – Aug 2023

Software Development Engineer (Intern)

- Designed a CI/CD pipeline for on-device ML training and federated learning, automating Scikit-learn → ONNX / Kotlin weight conversion and prototyping a Flower-based federated server for distributed Android devices.

MSD Aug 2022 – Dec 2022

Data Analytics Consultant (Apprentice)

- Led a team of 4 to build an internal Adverse Event tracking and drug benchmarking tool, shipping Power BI dashboards over a Python ETL and cloud ingestion pipeline for real-time clinical reporting.

Panasonic R&D Center May 2022 – Aug 2022

AI Algorithm Engineer (Intern)

- Built a real-time monitoring system combining PaddleOCR / Tesseract with fuzzy search for noisy industrial environments, optimising the edge inference pipeline to cut latency by 30%.

Projects

Byline | *Vertex AI, Multi-Agent Systems, Python*

- Multi-user scheduled newsletter service on a multi-agent workflow (planner, researcher, writer, editor) coordinating tool-use across HN, Reddit, GitHub, ArXiv, Perplexity, Yahoo Finance, and web search; typeset as a retro broadsheet-style PDF newspaper. Live at byline.siddhantpathakk.com.

Clarion | *Anthropic SDK, MCP, Next.js*

- AI chief-of-staff for the inbox built on the Anthropic SDK and MCP, turning noisy email into a single prioritized briefing with streaming chat, cited answers, a structured daily digest, and three swappable personas (Developer / Manager / Finance). Built at Push to Prod 2026, Singapore.

TAGON | *PyTorch, NumPy, Python*

- Formulated Temporal Attention Graph-Optimized Network (TAGON) architecture, leveraging GNNs and temporal attention for dynamic user interaction adaptation in recommendation systems; improved relevance and timeliness by 20% across five real-world datasets, outperforming SOTA baselines on precision, recall, and user satisfaction.

Icarus AI | *LLMs, Explainability, Full-Stack*

- Built an explainable job-recommendation platform that uses candidate CVs and LLMs to deliver personalised career suggestions with reasoning traces for each match.
- Awarded **Third Place** at the NTU Deep Learning Week Hackathon 2024 (MLDA@EEE Club) among competing university teams.

Delphi | *GCNs, AWS*

- End-to-end misinformation-detection pipeline using Graph Convolutional Networks over Twitter propagation patterns with Word2Vec embeddings on AWS, reaching 95% accuracy.
- Awarded **Innovative Use of Natural Language Processing** at the NTU Deep Learning Week Hackathon 2022 (MLDA@EEE Club).

Technical Skills

Languages: Python, SQL, R, C++, Java, Bash
GenAI & LLMs: RAG, Agentic AI, MCP, LangGraph, Google ADK, vLLM
ML: PyTorch, TensorFlow, ONNX, Scikit-learn, MLX
Cloud & MLOps: AWS, GCP, Vercel, Cloudflare, Docker, GitHub Actions, Datadog, CloudFormation, Vertex AI, SageMaker